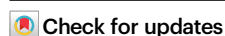


Real-time observation of the diffusion-limited formation of a cation-molecule complex

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For two molecules or atoms to react, they must first move towards each other and then, upon meeting, form new chemical bonds. Ultrafast spectroscopy and diffraction techniques have illuminated the bond-formation step for example by triggering reactions from weakly-bound precursors but not the initial approach, typically attributed to diffusion, although this is often the step that determines the reaction rate. Here, we measure and control the diffusion time for the reaction where a Li^+ ion forms a complex with a benzene dimer, a textbook cation- π system, inside a liquid helium nanodroplet. Using femtosecond-timed Coulomb explosion, we find that Li^+ , initially at the droplet surface over 30 Å from the dimer, first solvates, then diffuses ballistically at 43 m/s and finally reacts. These results, rationalized by ring-polymer molecular dynamics simulations, pave the way for real-time imaging of stereodynamics in ion-molecule reactions.

Unimolecular processes such as bond-breaking and isomerization start from the well-defined geometry of a single parent molecule. Therefore, their kinetics and even the changing molecular structure are, in principle, straightforward to explore on the natural atomic time scale using femtosecond (fs) pump-probe methods. This is not the case for a bimolecular reaction between two atomic or molecular reactants A and B, because such a process requires that A and B first meet. In gases and liquids, this happens via diffusion, and since the distance between reactants varies, so does the moment they meet for any defined starting time. Consequently, diffusion prevents the femtosecond time resolution needed to follow the part of the reaction where chemical bonds actually change. In certain cases, this timing problem may be overcome if a weakly-bound, non-reactive A-B complex can be created and if a fs pump pulse can initiate the bimolecular reaction by photoexciting A or B^{1–3}. Such a precursor complex eliminates diffusion and essentially turns the bimolecular reaction into a unimolecular process. Examples in gas phase and solvents have been reported, including proton transfer, vibrational dynamics and bond shortening^{4–8}. Another

approach, demonstrated for radical-molecule and photo-redox reactions in a solvent, is to increase the reactant concentration so much that the diffusion time becomes almost negligible^{9,10}. It has also been shown that the intrinsic rate constant of e.g. bimolecular electron transfer reactions in solution can be determined when theoretical modelling of diffusion is applied to fs time-resolved experimental signals¹¹.

Under most synthetic or natural conditions, the two reactants are, however, not in a close-proximity, precursor geometry. Rather, they are brought together via diffusion, which for many reaction types determines the overall reaction rate and may involve stereodynamical effects as the reactants approach each other. Insight into the progression of the entire bimolecular reaction therefore requires time-resolved techniques that also capture and possibly control the diffusion stage, a so far unmet challenge. Here, we introduce an experimental method that enables us to prepare two reactants, an ion and a molecule, at a controllable initial distance, large enough that diffusion must occur before they meet and react. The separation of the reactants

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is obtained using helium nanodroplets doped with an alkali atom, here Li, at its surface and a molecular system, here the benzene dimer Bz₂, near its center. A fs pump pulse initiates the reaction at a precise time through selective ionization of the Li atom. Coulomb explosion, triggered by ionizing Bz₂ with a delayed fs probe pulse, provides time-resolved information about both the initial solvation of the Li⁺ ion, its diffusion through the droplet and the final Li⁺Bz₂ complex formation. Ring-polymer molecular dynamics (RPMD) simulations serve to interpret the experimental findings and benchmark the experimentally determined diffusion velocity.

Results

Atomistic modelling of the Li⁺ + Bz₂ → Li⁺Bz₂ reaction in a helium nanodroplet

First, we provide a theoretical reference for the reaction:



obtained through RPMD simulations^{12,13}, details given in Methods and Supplementary Information. We consider a helium droplet containing 3600 atoms corresponding to a radius of 33 Å with Bz₂ near the center and a Li atom residing ~2 Å above the surface¹⁴. At time $t = 0$, the Li atom is instantly ionized. The ensuing dynamics, including the local environment of the Li⁺ ion, its distance to Bz₂ and the spatial orientation of Bz₂, are determined in real time. Figure 1a, illustrating the droplet and dopants at four selected times, shows that Li⁺ gradually moves towards the droplet center and eventually reaches Bz₂ to form a Li⁺Bz₂ complex. A movie of the RPMD trajectory is available as Supplementary Video 1.

The full curve in Fig. 1b, depicting the Li⁺–Bz₂ distance, $r(\text{Li}^+ \text{--} \text{Bz}_2)$, for a single trajectory, as a function of time, shows that Li⁺ initially accelerates into the droplet, then decelerates and at $t \sim 10$ ps settles at a constant velocity. These early dynamics, caused by the polarizability interaction between Li⁺ and the He atoms, see ref. 13, and Supplementary Information, are independent of the presence of Bz₂, as seen by comparing results without Bz₂ in Fig. 1e to those with Bz₂. The latter represents the average over 10 trajectories corresponding to different initial orientations of Bz₂ i.e., a situation more accurately reflecting the experimental conditions than the single trajectory in Fig. 1b. At longer times, Fig. 1b shows that the Li⁺ velocity remains constant until $t \sim 45$ ps, where the ion accelerates and $r(\text{Li}^+ \text{--} \text{Bz}_2)$ drops from 13 Å to 3.5 Å at $t = 52$ ps. Thereafter, $r(\text{Li}^+ \text{--} \text{Bz}_2)$ does not change. We ascribe this final step, only observed when Bz₂ is present, to the short-range Li⁺–Bz₂ attraction and the ensuing formation of the Li⁺Bz₂ complex, where Li⁺ binds to the outside of the dimer (Fig. 1a).

Figure 1c shows that the distance between the two Bz molecules, $r(\text{Bz} \text{--} \text{Bz})$, remains constant at ~3.9 Å, the calculated equilibrium distance of Bz₂, until $t = 51$ ps, where it drops to 3.2 Å within a few ps. This drop, highlighted by the grey shaded band, coincides with when Li⁺ makes the final approach to Bz₂ and is another manifestation of the Li⁺Bz₂ complex formation. Finally, Fig. 1d shows the coordination number for Li⁺, $N_c(\text{Li}^+)$, which is a measure of the number of He atoms closely bound to the ion. During the first ~4 ps, N_c increases from 0 to 6, an effect ascribed to the solvation of Li⁺^{13,14}. Thereafter, N_c increases slowly to about 8 and then, from $t = 51$ to 52 ps, it drops sharply to below 2 and subsequently recovers at 3. We interpret the sharp drop, occurring simultaneously with $r(\text{Bz} \text{--} \text{Bz})$ and $r(\text{Li}^+ \text{--} \text{Bz}_2)$ reaching their final values, as part of the solvation shell of Li⁺ being broken during its bonding to Bz₂. The asymptotic value of 3 for N_c is the number of He atoms attached to the side of Li⁺ facing away from Bz₂.

Initial solvation of the Li⁺ ion

The starting point for the experiments is also helium nanodroplets, each containing one Li atom, at the surface, and one benzene dimer, expected to be located near the droplet center. For experimental

reasons, many of the droplets will, however, contain a Li atom and only a single benzene (Bz) molecule (see Methods). First, the doped droplets are irradiated by a 50-fs pump laser pulse that multiphoton ionizes Li and leaves Bz₂ (or Bz) intact. This prompt creation of a Li⁺ ion sets a precise starting time for the reaction, Eq. (1), at a well-defined initial separation of the reactants determined by the radius of the droplets.

Next, the 60-fs probe pulse, sent at time t , ionizes Bz₂ (or Bz), which leads to ejection of different ion species, notably Li⁺, Bz⁺ and Li⁺Bz, from the droplet. We detect the time-dependent velocity-resolved yields of these ions to obtain information about different stages of the reaction path. First, we focus on the Li⁺ ion yield. Figure 2a–f shows the covariance maps^{15,16} between the distributions of $v(\text{Li}^+)$ and $v(\text{Bz}^+)$ at six selected times, $v(\text{Li}^+)$ and $v(\text{Bz}^+)$ denoting the velocities of Li⁺ and Bz⁺, respectively.

The pronounced covariance signal centred at (3.0 km/s, 0.2 km/s), observed at $t = 0.2$ ps, shows that a Li⁺ ion with v around 3.0 km/s, corresponding to a kinetic energy, E_{kin} , of 0.33 eV, is correlated with a Bz⁺ ion with v around 0.2 km/s ($E_{\text{kin}} \sim 0.02$ eV). This correlated ion-pair stems from the Coulomb repulsion between a Li⁺ and a Bz⁺ ion originating from the same droplet with Bz⁺ produced by dissociative ionization of Bz₂ or by ionization of Bz. In either case, if Bz₂ or Bz is located in the center of the droplets, with an expected average radius $\langle R \rangle$ of 35 Å, there will be a Coulomb energy of 0.37 eV between Li⁺ and Bz⁺, the moment Bz⁺ is created. Their repulsion causes the ejection of both ions from the droplet and momentum conservation gives the Li⁺ ion a final kinetic energy of 0.32 eV. The match with the measured E_{kin} corroborates that the initial Li⁺–Bz₂–separation is to a good approximation given by the droplet radius. For details, see Supplementary Information.

At $t = 0.6$ ps, Fig. 2b, the covariance signal at (3.0 km/s, 0.2 km/s) is weaker and at $t = 3.0$ ps, Fig. 2c, it has essentially disappeared. Figure 2g shows the Li⁺ signal, coincidence-filtered with Bz⁺ ions¹⁷, as a function of time. The signal drops exponentially to zero within the first three ps, which we ascribe to Li⁺ solvating in the droplet, similar to recent observations of solvation of a single Na⁺ ion in a He droplet^{18,19}. Details of the solvation dynamics of the Li⁺ ion are reported elsewhere²⁰.

Diffusion and bond formation

Our main interest here is what happens at longer times. Figure 2g shows that after the initial decrease, the Li⁺ signal recovers and rises gradually until it reaches a plateau around 200 ps. We believe the reappearance of the Li⁺ signal occurs when the solvated Li⁺ ion moves sufficiently close to Bz₂ (or Bz) that the probe-pulse-induced Coulomb repulsion strips off all He atoms in the solvation shell. The significantly higher $v(\text{Bz}^+)$ detected at longer times, Fig. 2d–f, compared to $v(\text{Bz}^+)$ for $t < 1$ ps, Fig. 2a–b, corroborates that Li⁺ has moved closer to Bz₂ (or Bz). Thus, the results in Fig. 2 qualitatively show that Li⁺ gradually approaches and probably reacts with Bz₂ (or Bz), but it is difficult to quantitatively determine when the Li⁺Bz₂ complex forms.

Therefore, we turn to another experimental observable, namely Li⁺Bz ions detected in coincidence with Bz⁺ ions. This observable, termed Co(Li⁺Bz, Bz⁺), has two decisive advantages. Firstly, unlike the signals shown in Fig. 2, it only originates from droplets doped with a Li atom and a Bz dimer. Secondly, Li⁺Bz is only formed when Li⁺ has moved close to Bz₂. Although this appears intuitively clear, we modelled the probe process and showed that formation of a Li⁺Bz–Bz⁺ ion pair only happens when $r(\text{Li}^+ \text{--} \text{Bz}_2) < 8$ Å (Supplementary Information). According to Fig. 1b, $r(\text{Li}^+ \text{--} \text{Bz}_2)$ decreases from 8 Å to 3.5 Å, the value of the final Li⁺Bz₂ complex, within ~1 ps. Thus, we have an experimental observable that is nonzero only when Li⁺Bz₂ has formed—or in the last ps before this happens.

Figure 3a displays the velocity distribution of the Li⁺Bz ions detected in coincidence with a Bz⁺ ion as a function of time. The figure

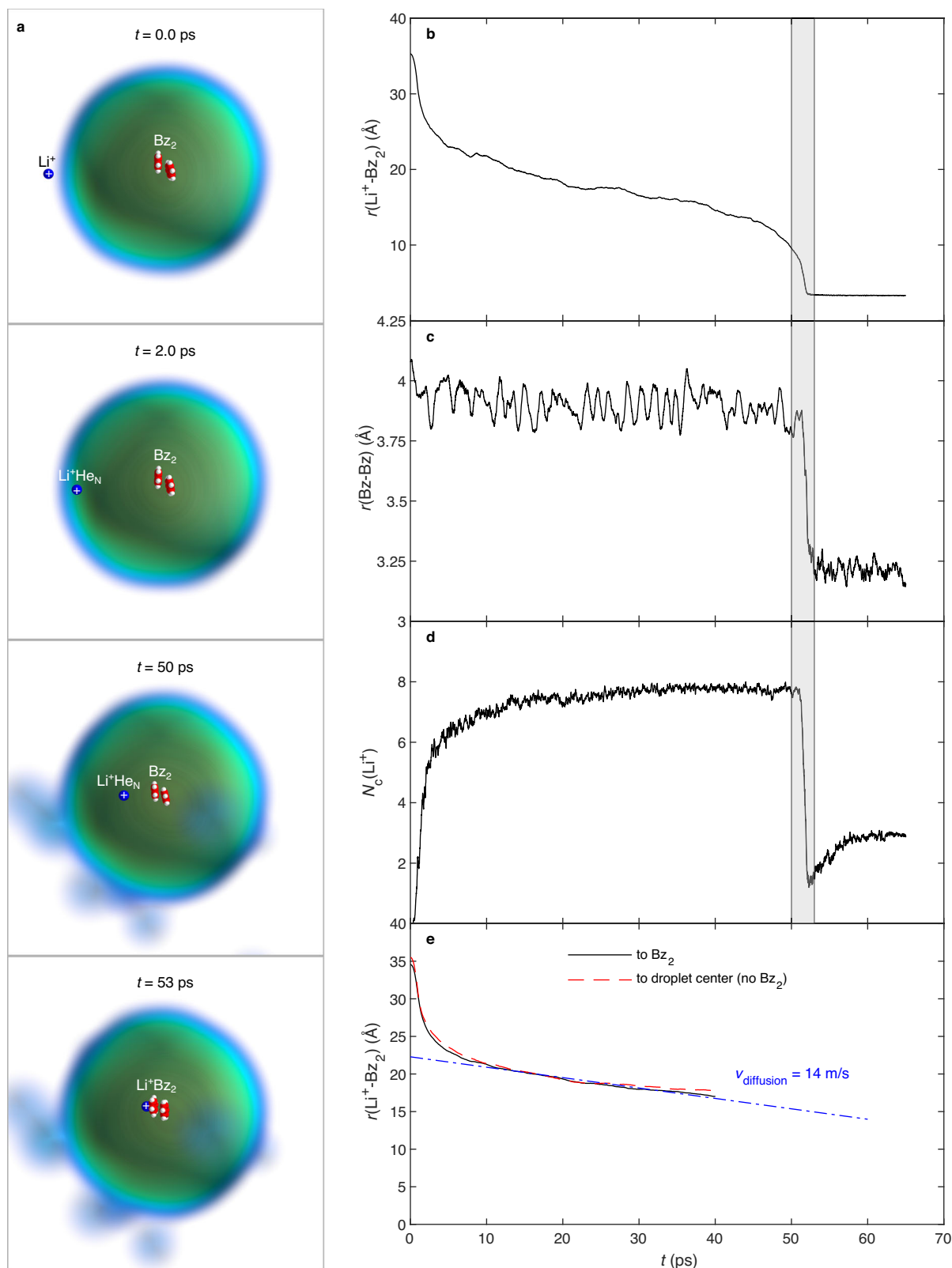


Fig. 1 | Ring-polymer molecular dynamics simulation of Li^+ solvating, diffusing and reacting with a benzene dimer. **a** Illustration of the Li^+ ion (blue circle) and Bz_2 (white and red shapes), both enlarged for visualization, inside a 3600-atom helium nanodroplet at four selected times after Li^+ is created. The droplet is shown as a continuous density medium after convoluting the He positions with a Gaussian broadening of 5 Å. In the lower two panels, the density outside the droplet region indicates helium evaporating^{14,18}. Time dependence of **b** Li^+-Bz_2 (center-of-mass) distance, **c** distance between the benzene molecules and **d** coordination number of

the Li^+ ion. The grey shaded band marks the interval where Li^+ binds to Bz_2 . **a–d** represent results from the same single RPMD trajectory. **e** Same as **b** but averaged over the trajectories from 10 independent simulations (due to computational constraints these trajectories were limited to $t = 40$ ps). The blue dash-dotted line, with a slope $v_{\text{diffusion}}$, is the best linear fit to the diffusion region, here $10 \text{ Å} < r(\text{Li}^+-\text{Bz}_2) < 40 \text{ Å}$. The red dashed curve shows the distance of Li^+ to the droplet center from the average of 10 independent simulations without Bz_2 . Source data are provided as a Source Data file.

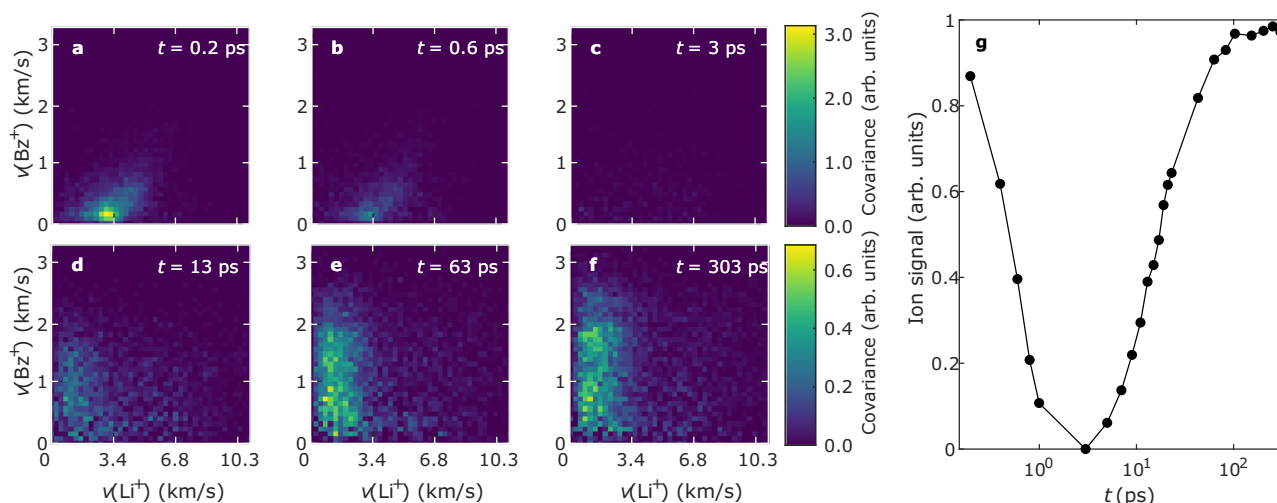


Fig. 2 | Dynamics of the initial solvation of Li^+ and of its subsequent complex formation with Bz or Bz_2 . **a–f** Covariance maps of the velocity distributions of Li^+ and Bz^+ ions at six selected times. **g** Filled circles: time dependence of the coincidence-filtered Li^+ ion signal, i.e., the Li^+ ions that, within the same laser shot,

are detected along with a Bz^+ ion having the opposite momentum. The full line connects the experimental points and serves to guide the eye. Note that the time scale is logarithmic. Details of the coincidence filtering are given in Methods. Source data are provided as a Source Data file.

shows that these ions first appear after about 5 ps and the distribution centres around 0.6 km/s, which we interpret as the final mean velocity of the Li^+Bz ions after Coulomb repulsion from a Bz^+ ion. Next, the time dependence of $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ is obtained by integrating the velocity distributions in Fig. 3a at each time step (Methods). The result, black circles in Fig. 3b, shows that the yield of Li^+Bz_2 complexes gradually increases from $t \sim 5$ ps until $t \sim 100$ ps where the signal flattens. This curve represents our measurement of the time-dependent yield of the reaction in Eq. (1).

To model and understand the experimental data, we note that according to the simulations, the total time, t_{total} , to form Li^+Bz_2 from the Li^+ and Bz_2 reactants at their initial separated positions, i.e., the time we measure, has three contributions: 1) The solvation time, $t_{\text{solvation}}$, of Li^+ including the initial acceleration and deceleration, i.e., the first ~ 10 ps in Fig. 1b; 2) The diffusion time, $t_{\text{diffusion}}$, where Li^+ moves with constant velocity, $v_{\text{diffusion}}$, i.e., ~ 10 –45 ps in Fig. 1b; 3) The bond formation time, t_{bond} , where Li^+ makes the final acceleration towards Bz_2 and forms Li^+Bz_2 , i.e., ~ 45 –52 ps in Fig. 1b. Thus,

$$t_{\text{total}} = t_{\text{solvation}} + t_{\text{diffusion}} + t_{\text{bond}} \quad (2)$$

Equation (2) is based on the simulation for a droplet with $R = 33$ Å, whereas the experimental results concern droplets with $33 \text{ Å} \leq \langle R \rangle \leq 49$ Å. Currently, it is not computationally feasible to simulate larger droplets, but we expect that for a limited increase of R , $t_{\text{solvation}}$, t_{bond} , and $v_{\text{diffusion}}$ remain essentially unchanged. Therefore, we rewrite Eq. (2) as

$$t_{\text{total}}(R) = R/v_{\text{diffusion}} + t_{\text{constant}} \quad (3)$$

and experimentally test its validity as described below.

The droplets are formed with a size distribution, well described by a log-normal distribution²¹. According to Eq. (3), we therefore expect a log-normal distribution of t_{total} . However, $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ represents integration over t_{total} , because the smallest droplets give a contribution at the earliest times and then the contributions from larger droplets gradually add at longer times. Thus, $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ should be given as the cumulative distribution function of t_{total} , denoted CDF_{LN} (see Methods). The best fit of CDF_{LN} to the data points, blue curve in Fig. 3b, closely matches the experimental results. We repeated the measurements for 14 other droplet size distributions and the results

for three of them are displayed in Fig. 3b–e. Again, CDF_{LN} provides excellent fits to the data points, and it is seen that the reaction time increases as the droplet size increases consistent with a diffusion-limited reaction.

From each fit, we obtain the mean value of the log-normal distribution of t_{total} . This value, denoted $\langle t_{\text{total}} \rangle$, represents the average reaction time at the particular droplet size distribution. Figure 3f displays $\langle t_{\text{total}} \rangle$ as a function of $\langle R \rangle$. It appears that for $33 \text{ Å} \leq \langle R \rangle \leq 42 \text{ Å}$, $\langle t_{\text{total}} \rangle$ scales linearly with $\langle R \rangle$. The linear dependence is quantified by the dashed, blue line representing the best linear fit to the points in the 33–42 Å range. The linearity validates Eq. (3) and allows a determination of the average velocity, $\langle v_{\text{diffusion}}^{\text{exp}} \rangle$, of Li^+ in the diffusion region, as the inverse of the slope of the fitted line. We find $\langle v_{\text{diffusion}}^{\text{exp}} \rangle = 43 \pm 5$ m/s (see Supplementary Information), i.e., larger than the simulated $v_{\text{diffusion}} = 14$ m/s, Fig. 1e.

Our simulations show that the linear motion results from the combined effect of Li^+ being accelerated by the attractive interaction with the He atoms and Bz_2 and decelerated by friction from the droplet (Supplementary Information). The simulations do, however, not account for the superfluidity that characterizes helium nanodroplets^{22,23} and therefore $v_{\text{diffusion}}$ is expected to underestimate the real diffusion velocity. The found $\langle v_{\text{diffusion}}^{\text{exp}} \rangle$ is close to the critical Landau velocity measured to be 56 m/s for Ag atoms in helium droplets²³, indicating that the onset of friction on Li^+ occurs when its speed is slightly below the Landau velocity. Similar to the findings in Ref.²³, we believe this is caused by heating of the droplet—in our case due to the ion solvation process. Finally, Fig. 3f shows that for the larger droplets, the reaction time gradually increases compared to the extrapolated linear behaviour. This could be caused by increased delocalization of Bz_2 away from the droplet center²⁴, meaning Li^+ not always moves straight towards Bz_2 , i.e., it may have to travel farther than the droplet radius to encounter the dimer. This effect is expected to be more pronounced for larger droplets in agreement with the observed behaviour of $\langle t_{\text{total}} \rangle$. The excellent agreement between the fits of CDF_{LN} and the experimental points suggests that it is unlikely that the delocalization becomes so pronounced that the Li^+ ion does not meet Bz_2 in the center – such an event would give a deviation from the log-normal distribution.

Discussion

Characterization of diffusion in fs time-resolved experiments of bimolecular reactions in classical liquids was addressed previously. For

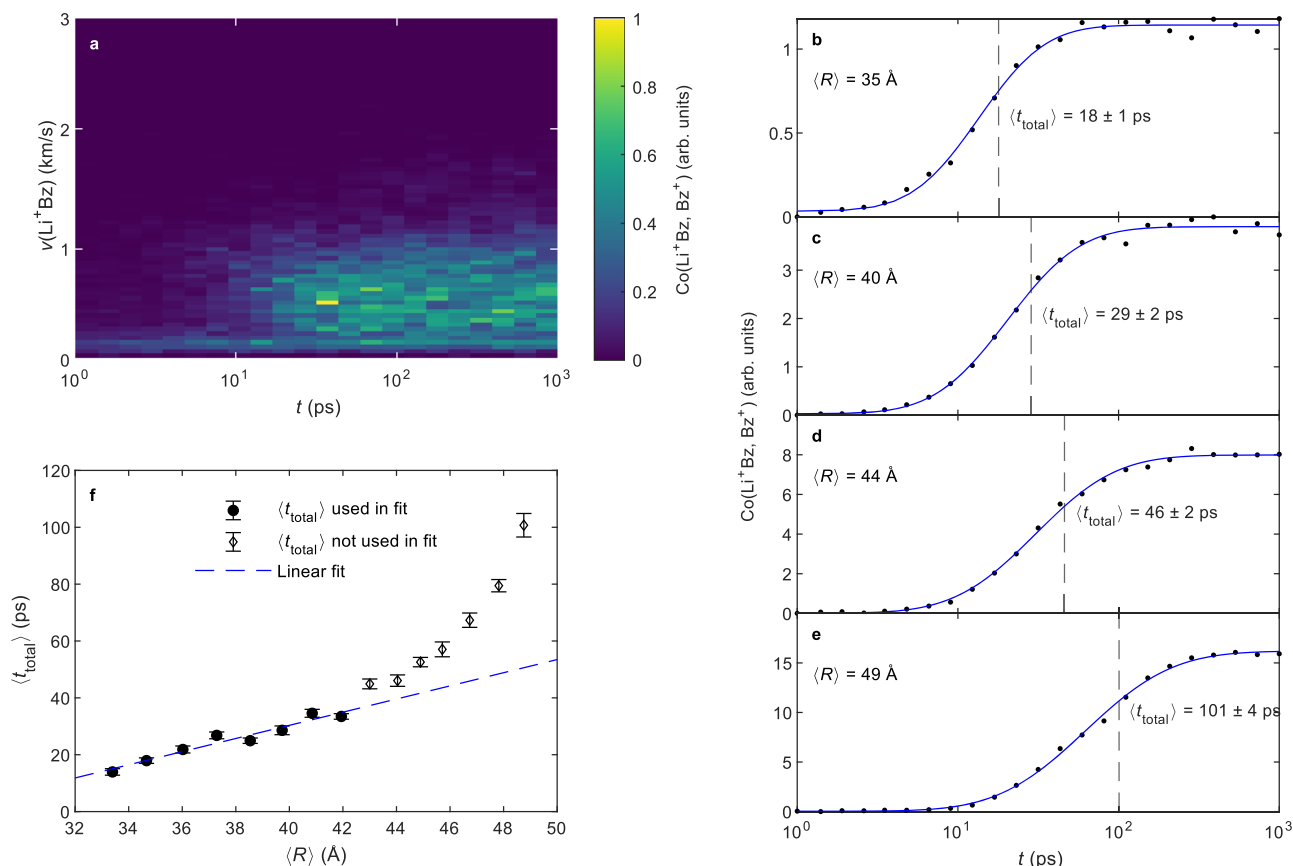


Fig. 3 | Time-dependent yield of the reaction $\text{Li}^+ + \text{Bz}_2 \rightarrow \text{Li}^+\text{Bz}_2$ for different droplet sizes, and determination of the Li^+ diffusion velocity. **a** Time dependence of the velocity distribution of Li^+Bz ions coincident with a Bz^+ partner ion. The average droplet size $\langle R \rangle$ is 35 Å. **b–e** Black dots: $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ as a function of time for different $\langle R \rangle$ indicated on each panel. Blue curves: Best fit of the cumulative log-normal distribution function CDF_{LN} to the data points. The dashed line shows

the mean value, $\langle t_{\text{total}} \rangle$, of the log-normal distribution of the total reaction time, t_{total} . The uncertainty is calculated by propagating the 1σ uncertainties of μ_t and σ_t obtained by the fitting procedure of CDF_{LN} . **f** Points: $\langle t_{\text{total}} \rangle$ measured at 15 different $\langle R \rangle$. Dashed line: Best linear fit to the data points included. Methods includes an explanation of how $\langle R \rangle$ is estimated. Source data are provided as a Source Data file.

instance, transient absorption spectroscopy examined reactions between a chlorine radical and an alkane solvated in dichloromethane²⁵. Smoluchowski diffusion modelling showed the reactions were diffusion-limited with rate constants of about $1 \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$. In comparison, for the diffusion-limited reaction explored here, we estimate the reaction constant based on the reaction time for a given droplet size and find $5 \times 10^{12} \text{ M}^{-1} \text{ s}^{-1}$ (Supplementary Information). Thus, diffusion of the Li^+ ion in the He droplet is two orders of magnitude faster. This difference reflects two qualitatively different types of diffusion. In classical liquids, diffusion arises from Brownian motion, with diffusion time proportional to the mean square displacement²⁶. In contrast, Li^+ moves with constant velocity through the droplet, and diffusion time depends linearly on distance, see Eq. (3). The linear motion is an example of anomalous diffusion termed ballistic diffusion^{27–29}, a phenomenon also encountered under widely different conditions, for instance in the transport of molecules in DNA model condensates³⁰.

Currently, the time resolution of our technique is insufficient to capture details of the part of the reaction where new molecular bonds form. To explore this, it may be favourable to start from close-proximity precursor complexes. Helium droplets offer opportunities in this regard by enabling formation of diverse molecular and atomic complexes^{31,32}, going beyond what can be accomplished using gas-phase samples or molecules in classical solvents. Starting from a close-proximity precursor misses, however, the stereodynamics of reactions, i.e., as two reactants approach, their mutual interaction may steer their relative

orientation into a particular reaction geometry^{33–36}. To explore this fundamental aspect of bimolecular reactions, the two reactants should start far apart, as ensured by the present method. In this connection, we point out that stereodynamics of bimolecular reactions have been explored extensively using crossed molecular beams^{37,38}. In those works, typically addressing inelastic or reactive scattering between an atom and a molecule, the molecule was aligned or oriented along a laboratory-fixed axis by a static electric field sometimes preceded by quantum-state-selection or photo-selection. Varying the molecular orientation or alignment with respect to the direction of the incoming atom and recording the total or differential cross section have provided deep insight into how steric effects influence chemical reactions.

Our method can address stereodynamics in two fundamentally different ways. Firstly, the molecule is not aligned or oriented by an external field defined in the laboratory frame. Instead, it is the approaching ion that induces alignment or orientation of the molecule through the interaction between its electrical properties and the electric field from the cation, see Supplementary Video 2. Secondly, our experimental method is time-resolved. In the Li^+-Bz_2 reaction, Bz_2 gradually aligns the normal of its plane towards the trajectory of the incoming Li^+ ion, (Supplementary Information). Since it has already been demonstrated that laser-induced Coulomb explosion can determine structure and alignment of molecules and molecular complexes inside He droplets^{39,40}, we believe that our technique can be extended to provide real-time imaging of stereodynamics in ion-molecule reactions^{41–44}.

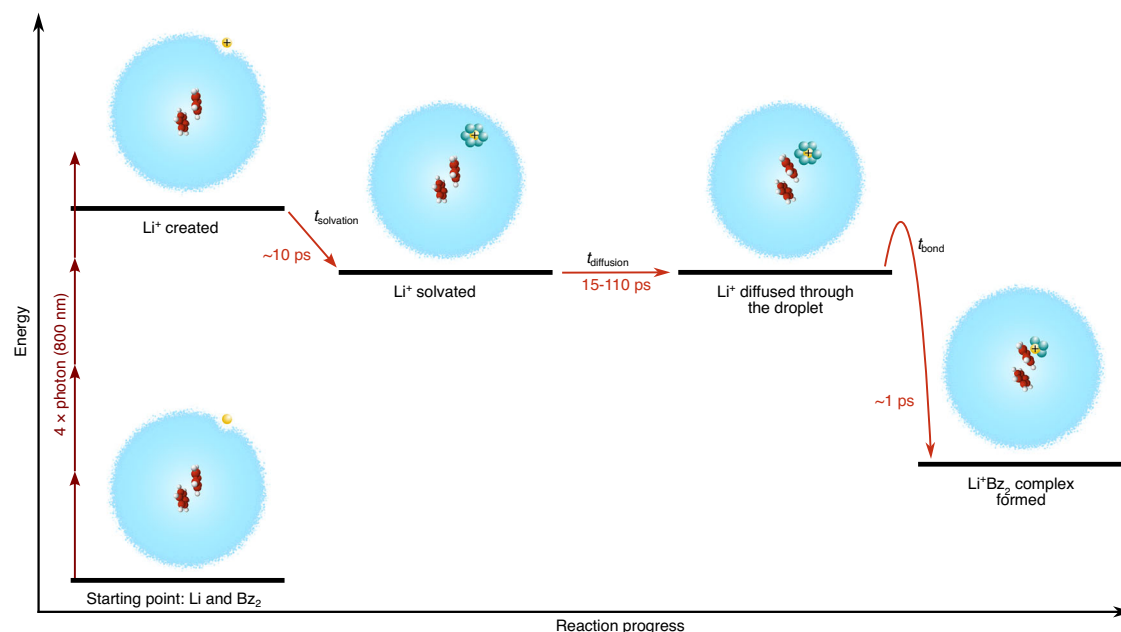


Fig. 4 | Reaction pathway for $\text{Li}^+ + \text{Bz}_2 \rightarrow \text{Li}^+\text{Bz}_2$. Upon sudden formation, by the pump-pulse-induced 4-photon ionization of a Li atom at the droplet surface, the Li^+ ion solvates, then diffuses ballistically with a velocity of 43 m/s and finally binds to Bz_2 near the droplet center to form Li^+Bz_2 . In addition to the timescales of solvation, diffusion and bond-formation, the figure illustrates that the energy is lowered

by ~ 0.5 eV as Li^+ solvates¹⁴ and by another ~ 1.7 eV when it binds to Bz_2 (Supplementary Information). The latter number expresses that the noncovalent cation- π interaction is almost as strong as a typical covalent bond. Energy levels and atomic sizes are not to scale.

In conclusion, we have introduced an experimental method to explore, in real time, the bimolecular reaction where a Li^+ ion diffuses towards a benzene dimer and forms a Li^+Bz_2 complex inside a liquid helium nanodroplet. In combination with atomistic RPMD simulations, the experimental results allow us to draw a picture of the progression of the entire reaction summarized in Fig. 4. The Li^+Bz_2 complex is an example of an ion-molecule system held together by noncovalent bonding referred to as cation- π interaction⁴⁵. This interaction, which has not been explored previously by fs time-resolved techniques, plays an important role for e.g., molecular recognition in biological systems and in organic synthesis⁴⁶. One opportunity opened by our work is to measure the stereodynamics of the approaching reactants using timed Coulomb explosion imaging. This should be possible for a variety of molecules reacting with atomic or molecular ions of alkali and alkaline earth atoms. Another opportunity is to explore interactions between alkali atoms or dimers at the droplet surface and fullerenes in the interior. The current method may provide insight into electron transfer in such systems – recently explored by electron ionization mass spectrometry⁴⁷, helium density functional theory⁴⁸ and path-integral molecular dynamics simulations^{49,50}.

Methods

Experimental details

An illustration of the experimental setup is shown in Supplementary Information. More details can be found in Ref. 19. A beam of liquid helium droplets is formed by subcritical expansion of 99.9999% purity ^4He gas into vacuum through a $5\text{ }\mu\text{m}$ diameter orifice. The backing pressure of the He gas was 50 bar in all experiments, while the temperature of the expansion nozzle was varied between 13.5 K–20.5 K. The resulting droplet distributions had an average number of helium atoms between 3200 and 14600. The formed helium droplet beam passes through two doping cells. The first contains a vapor of benzene molecules, let into the cell through a leak valve. The second contains a gas of Li atoms obtained by heating a sample of solid Li in an oven. The set-point of the PID controller was set to 390 °C. The opening of the leak valve and the temperature of the Li metal were selected to

maximize the likelihood of doubly doping with benzene and singly doping with Li. The doping process is Poissonian²², so other combinations of dopants are unavoidable. The case where a droplet is doped with one Bz molecule and one Li atom is addressed in the main text, and while other systems like LiBz_3 , Li_2Bz , Li_2Bz_2 etc. are possibly formed, they are present to a much lesser extent and do not contribute noticeably to the signal.

The doped droplet beam enters the target vacuum chamber, that contains a velocity map imaging (VMI) spectrometer^{51,52}. The VMI spectrometer is a three-plate setup, where the repeller (6000 V), extractor (3790 V) and ground (0 V) plate voltages are chosen to achieve velocity focusing. The droplets are irradiated by two pulsed, linearly polarized laser beams in the center of the VMI spectrometer. The two beams are spatially overlapped, and their linear polarization is in the plane of the detector at the end of the VMI spectrometer. The laser pulses are produced by a Coherent Astrella HE laser system (wavelength (nominal) $\lambda_0 = 800$ nm, temporal FWHM $\tau = 35$ fs, pulse energy $E_p = 2$ mJ, repetition rate $f_{\text{rep}} = 5$ kHz). The focused pump pulses used to ionize Li ($\tau = 50$ fs, $\lambda_0 = 800$ nm, $E_p = 16$ μJ , beam waist $w_0 = 41$ μm) have a peak intensity of $I_0 = 1.1 \times 10^{13}$ W/cm², while the probe pulses used to ionize Bz ($\tau = 50$ fs, $\lambda_0 = 400$ nm, $E_p = 18$ μJ , $w_0 = 28$ μm) have a peak intensity of $I_0 = 2.7 \times 10^{13}$ W/cm². The laser parameters were chosen to selectively ionize Li with the pump pulse and efficiently ionize Bz with the probe pulse. The probe pulse was delayed by time t using a linear delay stage.

The ions produced are projected by the VMI spectrometer onto a two-dimensional imaging detector. The detector consists of two micro-channel plates (MCP) and a phosphor screen (P47). The first MCP is a funnel-type plate, which increases the detection efficiency⁵³. The front of the MCP stack was grounded, while the back was set to 2300 V. The middle of the stack was kept at 1150 V using a shim-ring and the phosphor screen was biased to 4900 V. The light produced by the phosphor screen was imaged by a TPX3Cam detector^{54–56}. The TPX3Cam is event-based and records both the position and time-of-arrival of each lit-up pixel independently, which allows us to measure VMI images of multiple, different ions. The pixel data stream is

time-walk corrected^{57,58}, clustered and centroided using the DBSCAN algorithm, and time-walk corrected again to retrieve the time-of-flight (ToF) from the time-of-arrival. The x - and y -hit coordinates are calibrated to give the projected velocities, v_x and v_y , using velocities of Li^+ ions coming from the Coulomb explosion of the $^3\Sigma_u^+$ state of the Li_2 dimer⁵⁹. Mass-over-charge values (m/q) were obtained by calibrating the ToF-spectrum using known background ions (H_2O^+ and effusive Bz^+). The final data stream then consists of tuples of values pertaining to each ion, including the laser shot number, ($ls, m/q, v_x, v_y$). The experiment was recorded for 15 different nozzle temperatures with a set of time delays between 1 ps and 1000 ps, where a trace was repeated between five and 20 times to achieve a reasonable signal-to-noise ratio.

Data analysis for extraction of $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$

The main observable in the experiment is the yield of Li^+Bz ions detected in coincidence with Bz^+ ions, denoted $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$. The experimental data acquired in the experiment consists of tuples of values from the TPX3Cam containing the shot number, mass-over-charge, velocity x -component and velocity y -component, ($ls, m/q, v_x, v_y$). The data analysis employed to extract $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ from these tuples is shown in Supplementary Fig. 10, for a dataset recorded with $\langle R \rangle = 35 \text{ \AA}$. First, Li^+Bz ions and Bz^+ ions are assigned in the m/q -spectrum, Supplementary Fig. 10a. VMI images are then constructed by calculating the 2D (v_x, v_y) histogram of the selected ion hits across all laser shots. A VMI image of Li^+Bz is shown in Supplementary Fig. 10b for a time-resolved experiment, where images from each time delay are summed. The image shows a strong circular feature with a radius of around $\sim 0.4 \text{ km/s}$. There is also a dark line in the center of the image, which is attributed to a burn-in on the MCP detector. The VMI image shows that Li^+Bz ions indeed are formed in the experiment and the majority of these have very low kinetic energy.

To gain further information about the time-resolved dynamics, we employ covariance map analysis^{15,16,60,61} of the velocities and recoil directions of Li^+Bz and Bz^+ ions. Covariance maps between the radial-velocity distributions, $v(\text{Li}^+\text{Bz})$ and $v(\text{Bz}^+)$, of the corresponding VMI images are shown in Supplementary Fig. 10c, similarly summed across a time-resolved experiment. The covariance map shows that Li^+Bz ions can be observed in coincidence with Bz^+ ions and that these ions have a well-defined kinetic energy that is larger than the dominating background signal observed in the VMI image in Supplementary Fig. 10b. To further corroborate this analysis, we also consider the covariance map between the angular distributions, $\theta_{\text{Li}^+\text{Bz}}$ and θ_{Bz^+} , shown in Supplementary Fig. 10d. The covariance map shows a 180° angular relation between the two ions, which is characteristic of a Coulomb explosion between two charges⁶². The angular covariance map is very wide compared to gas-phase experiments⁶², which is attributed to non-axial recoil effects coming from collisions with He atoms inside the droplet as the ions leave. The joint analysis of the velocities and recoil directions in Supplementary Fig. 10c-d shows that Li^+Bz ions with a relation to Bz^+ ions representative of a Coulomb explosion process can be observed and that these ions have a kinetic energy that is larger than what can be seen in the VMI image of Li^+Bz , see Supplementary Fig. 10b. The covariance maps are then used to coincidence filter the Li^+Bz ions¹⁷. The filtering parameters used are:

$$-120^\circ < |\theta_{\text{Li}^+\text{Bz}} - \theta_{\text{Bz}^+}| - 180^\circ < 120^\circ \quad (4)$$

$$|\nu(\text{Li}^+\text{Bz}) - 0.50 \cdot \nu(\text{Bz}^+)| < 0.8 \frac{\text{km}}{\text{s}} \quad (5)$$

The tolerances are chosen to fully cover the covariance maps shown in Supplementary Fig. 10c-d. The relation between the two velocities, $\nu(\text{Li}^+\text{Bz})$ and $\nu(\text{Bz}^+)$, is not defined by the mass ratio between the species but is found empirically. This discrepancy is attributed to

the ions having different interactions with the helium droplet as they leave the droplet. The filtered VMI image is shown in Supplementary Fig. 10e for Li^+Bz . There is still a significant signal in the center of the image, but a second channel now appears in the image with a radius of $\sim 0.8 \text{ km/s}$. To analyze this new feature, we extract the full velocity distribution, $P(v_r)$. The velocities reported so far have been the projected velocities in the plane of the detector, v_x and v_y , but the full velocity distribution can be reconstructed by Abel inverting the image, here using the MEVIR algorithm⁶³. The normalized $P(v_r)$ distributions for the raw and filtered VMI images are shown in Supplementary Fig. 10f. By applying coincidence filtering, a broad peak in the velocity distribution is revealed, which is suppressed by background signal in the raw image. This peak is attributed to the Coulomb explosion between Li^+Bz and Bz^+ ions inside the droplet. $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ is then extracted by radially integrating the velocity distributions. In the experiment, this analysis is performed for each time delay. The resulting $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ trace for $\langle R \rangle = 35 \text{ \AA}$ is shown in Supplementary Fig. 10g and is identical to Fig. 3b. The panel also includes a fit of the cumulative-distribution function of a log-normal distribution, CDF_{LN} . The cumulative-distribution function is defined as:

$$\text{CDF}_{\text{LN}}(t) = \frac{1}{2}A \left[1 + \text{erf} \left(\frac{\ln(t) - \mu_t}{\sigma_t \sqrt{2}} \right) \right] + c \quad (6)$$

where A , c , μ_t and σ_t are fitted to $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$. A and c are parameters used to scale and offset the distribution. $\langle t_{\text{total}} \rangle$ can then be calculated as the mean of the log-normal distribution:

$$\langle t_{\text{total}} \rangle = \exp \left(\mu_t + \frac{\sigma_t^2}{2} \right) \quad (7)$$

Theoretical details of ring-polymer molecular dynamics

Simulations of the ionization, diffusion and reaction processes were conducted at the atomistic level of detail, using the same approach previously implemented for the alkali-xenon pair of dopants inside helium nanodroplets¹⁹. The system is prepared in the neutral state with the alkali atom sitting outside the droplet and the molecular dopant Bz_2 sitting at the center of the droplet. Thermostatted path-integral molecular dynamics (PIMD) were carried out before sudden ionization of the lithium atom takes place, at a temperature of 2 K and using time a step of 0.5 fs and a Trotter discretization number of 32, which is sufficient to ensure a liquid droplet. After 10 ps of equilibration, ring-polymer molecular dynamics (RPMD) simulations²⁰ were subsequently integrated up to 50 ps, using the same time step of 0.5 fs but without thermostating, assuming the potential energy surface now corresponds to Li^+ . The details of the interaction potentials for the neutral and cationic systems are described in the Supplementary Information. Different initial conditions for the trajectories were generated by supplementing the potential used in the equilibrated PIMD simulations with a simple biasing potential, detailed in the Supplementary Information. The dynamics of the $\text{Li}^+\text{-Bz}_2$ complex formation process was characterized using several descriptors that were monitored as a function of time, namely the distance $r(\text{Li}^+\text{-Bz}_2)$ between the lithium cation and the Bz dimer, instantaneously measured at the dimers center of mass, the distance $r(\text{Bz-Bz})$ between the two benzene molecules, instantaneously measured as the minimum distance between all atomic pairs of the two different molecules, and the coordination number $N_c(\text{Li}^+)$ of helium atoms around Li^+ , defined by summing a cut-off function over all $\text{Li}^+\text{-He}$ pairs with value 1 for distances below $3.8 \text{ \AA}$, 0 for distances above $4.2 \text{ \AA}$, and smoothly interpolated between these values using a fifth-order polynomial. These values were chosen to yield a strictly vanishing coordination number for the neutral lithium atom sitting above the droplet. All the structural

properties above were further averaged over the individual real-space beads in the path-integral description.

Estimation of mean droplet radius $\langle R \rangle$

A crucial parameter used in the analysis to determine the diffusion velocity of Li^+ in liquid helium droplets is the mean droplet radius, $\langle R \rangle$. Direct measurements of the helium droplet size distributions typically require a dedicated experimental setup^{21,64} and is therefore not currently possible in our experiment. Experimentally recorded droplet sizes have, however, been reported for conditions similar to ours also using a $5\text{ }\mu\text{m}$ orifice and similar stagnation pressures and temperatures²². These measurements form the basis for estimating the size of the droplets used in our experiment. To this aim, we first consider an undoped droplet size distribution characterized by a log-normal distribution⁶⁵:

$$P(N) = \frac{1}{\sqrt{2\pi N \delta_N}} \exp\left(-\frac{(\ln N - \mu_N)^2}{2\delta_N^2}\right) \quad (8)$$

where N is the number of helium atoms and δ_N and μ_N are parameters that characterize the distribution. For the expansion conditions in our experiment, the two parameters are given as⁶⁵:

$$\delta_N = 0.55 \quad (9)$$

$$\mu_N = \ln(\langle N \rangle) - \frac{\delta_N^2}{2} \quad (10)$$

which are uniquely determined by the mean number of helium atoms, $\langle N \rangle$. The mean radius, $\langle R \rangle$, can then be found assuming that the droplets are hard spheres with uniform density as:

$$\langle R \rangle = 2.22\text{ }\text{\AA} \cdot \langle N \rangle^{\frac{1}{3}} \exp\left(-\frac{\delta_N^2}{3}\right)^{\frac{1}{3}} \quad (11)$$

Estimated values for $\langle N \rangle$ and $\langle R \rangle$, based on previously reported droplet sizes²², are shown in Supplementary Table 2 for the nozzle temperatures used in the experiment. The droplets of interest in our experiment, which give rise to our observable $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$, are those that are doubly doped with benzene molecules and singly doped with lithium atoms. To estimate the size distributions of these droplets, we must then also include the doping probability of the molecules and atoms. The doped droplet size distribution is determined as the renormalized product of the two distributions:

$$P(N; k) = \frac{P(N) \cdot P(k; N, p_d)}{\int_0^\infty P(N') \cdot P(k; N', p_d) dN'} \quad (12)$$

where $P(k; N, p_d)$ is the probability of doping k particles at pressure p_d . This probability is described by a Poissonian distribution:

$$P(k; N, p_d) = \frac{(\beta_N p_d)^k}{k!} \exp(-\beta_N p_d) \quad (13)$$

where k is the number of doped particles, p_d is the doping pressure, N is the number of He atoms and β_N is a collection of factors:

$$\beta_N = F_{a0} \left(\infty, \frac{v_D}{v_d} \right) \cdot L_d \cdot \frac{\sigma_D(N)}{k_B T_d} \quad (14)$$

Here, L_d is the length of the doping cell, $\sigma_D(N)$ is the cross-section of the droplets, k_B is the Boltzmann constant and T_d is the dopant temperature. The correction factor F_{a0} accounts for the movement of the droplets and dopant particles, and includes the most probable

speed of the dopants ($\widehat{v}_d$) and the speed of the droplets (v_D):

$$F_{a0} \left(\infty, \frac{v_D}{v_d} \right) = \frac{1}{\sqrt{\pi}} \left(\frac{\widehat{v}_d}{v_D} \exp\left(-\left(\frac{v_D}{v_d}\right)^2\right) + \left(2 + \left(\frac{v_D}{v_d}\right)^2\right) \frac{\sqrt{\pi}}{2} \text{erf}\left(\frac{v_D}{v_d}\right) \right) \quad (15)$$

with $v_D = \sqrt{\frac{5T_{\text{nozzle}}k_B}{m_{\text{He}}}}$, $\widehat{v}_d = \sqrt{\frac{2T_d k_B}{m_d}}$ and $\text{erf}(x)$ the Gaussian error function. The temperatures of the dopants were $T_{\text{Bz}} = 298\text{ K}$ and $T_{\text{Li}} = 628\text{ K}$ and the doping cells had lengths of $L_{\text{Bz}} = 7.7\text{ cm}$ and $L_{\text{Li}} = 1.7\text{ cm}$. Based on the experimental data given in reference⁶⁵, the droplet cross section can be written as:

$$\sigma_D(N) = \pi \cdot \left(2.22\text{ }\text{\AA} \cdot N^{\frac{1}{3}} + 6.51\text{ }\text{\AA}\right)^2 \quad (16)$$

The last parameters required to estimate the doped droplet distribution are the vapor pressures of the samples. The optimal vapor pressures of benzene and lithium were found in the following way. We performed a pump-probe experiment at $t = 500\text{ ps}$, where $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ was maximized by varying the leak valve opening for $T_{\text{nozzle}} = 17\text{ K}$, see Supplementary Fig. 11. This leak valve position was then used throughout the experiment for other nozzle temperatures. The maximum of $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ must coincide with the vapor pressure of benzene where the doping of two molecules is most likely. This vapor pressure was then found by maximizing the overlap of the two distributions, $\int_0^\infty P(N) \cdot P(k; N, p_d) dN$, as a function of p_d . The vapor pressure was found to be $p_{\text{Bz}} = 1.2 \cdot 10^{-5}\text{ mbar}$. The vapor pressure of the Li sample was similarly estimated as the pressure which optimizes the likelihood of doping one lithium atom for the 17 K droplet size distribution, $p_{\text{Li}} = 1.99 \cdot 10^{-5}\text{ mbar}$, which corresponds to a temperature of $T_{\text{Li}} = 628\text{ K}$ ⁶⁶. Note that this is not entirely consistent with the vapor pressure of Li metal at the set temperature of the oven, which indicates that the temperature measured on the outside of the oven is not exactly the same as the temperature of the sample. The final doped droplet distribution was then found by first estimating the undoped droplet distribution, using values in Supplementary Table 2, finding the doped droplet distribution for $k = 2$ with benzene and then finding the doubly doped droplet distribution for $k = 1$ with lithium. The resulting droplet size distribution was then transformed from N to R and the numerical average $\langle R \rangle$ was computed. The effect of including the doping probabilities on $\langle R \rangle$ is shown in Supplementary Fig. 12 for the experimental conditions along with the undoped distributions.

Data availability

The data generated in this study are provided in the Supplementary Information/Source Data file. Source data are provided with this paper and are available at (<https://doi.org/10.6084/m9.figshare.30344179>). Source data are provided with this paper.

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Author contributions

J.K.C., C.E.P., S.H.A. and J.G. designed and built the experimental set-ups, maintained the experimental apparatus and carried out the experiments. J.K.C., C.E.P. and S.H.A. and H.S. analyzed the experimental data. F.C. designed, wrote the code for and carried out the molecular dynamics simulations and all electronic structure calculations. All authors discussed and interpreted the data. H.S., J.K.C. and F.C. wrote the manuscript and supplementary text and all authors contributed to the editing and reviewing.

Competing interests

The authors declare that they have no competing interests.

Additional information

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